Date:

Software Requirement Specification (SRS)

PSB Hackathon 2026

Software Requirement Specification
Version 1

Project Name: **Quantum-Proof Systems Scanner**

Team Name: **Spand**

Institute Name: **GEC Madhubani**

Revision History

Version No	***Date***	***Prepared by/Modified by***	***Significant Changes***
Draft V1.0			

Declaration

The purpose of this Software Requirements Specification (SRS) document is to identify and document the user requirements for the Quantum-Proof Systems Scanner The end deliverable software that will be supplied by ***Spand*** will comprise of all the requirements documented in the current document and will be operated in the manner specified in the document. The Source code will be developed subsequently based on these requirements and will formally go through code review during testing process.

Mentor Details :

Institute Name	GEC Madhubani
Signature: Date: 12/03/26	abuzar
Name & Title	md. abuzar, professor

Team Member Details:

Member 1	***Member 2***	***Member 3***	***Member 4***
Institute Name:	***Institute Name:***	***Institute Name:***	***Institute Name:***
gec madhubani Signature: Ravi	***gec madhubani*** Signature: purnima	***gec madhubani*** Signature: muskan	***gec madhubani*** Signature: gautam
Date: 13/03/26 Name: Ravi choudhary	Date: 13/03/26 Name: purnima kumari	Date: 13/03/26 Name: muskan kumari	Date: 13/03/26 Name: Gautam singhal
Team Lead	***Developer***	***Tester***	design research

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Software Requirement Specification (SRS)

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to identify and document the user requirement for Quantum-Proof Systems Scanner

This document is prepared with the following objectives:

To provide behaviour of the system

To provide Process Flow charts

1.2 Scope

Discover cryptographic inventory (TLS certificates, VPN endpoints, APIs).

Identify cryptographic controls (cipher suites, key exchange mechanisms, TLS versions).

Validate whether deployed algorithms are quantum-safe.

Generate actionable recommendations for non-PQC ready assets.

Issue digital labels: ***Quantum-Safe***, ***PQC Ready***, or ***Fully Quantum Safe***.

Enterprise wide console for Central management: A GUI console to display status of scanned systems (public facing applications) covering details mentioned in ***Appendix-A (Cert-In CBOM Elements).*** As per the variation of score (like High, Medium, Low rating etc) for any public applications, dashboard should display that change as well.

1.3Intended Audience

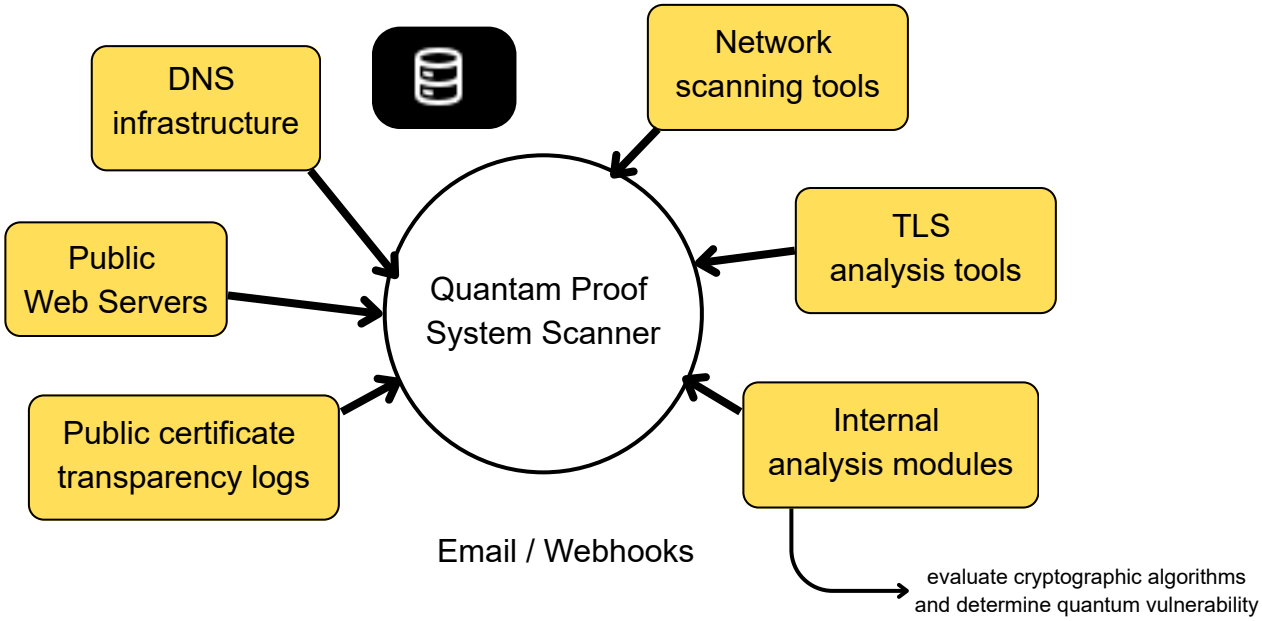
The intended audience of this document is business and technical users from PNB.

Overall Description

2.1 Product Perspective

The proposed system is a **Quantum-Safe Cryptographic Inventory Scanner** designed to identify and analyze cryptographic implementations used in public-facing applications such as web servers, APIs, and TLS-based services.
The system evaluates cryptographic implementations used in services such as web servers, APIs, email servers, file transfer services, and VPN endpoints. It extracts cryptographic configurations, evaluates their resistance to quantum attacks, and generates a Cryptographic Bill of Materials (CBOM) to assist organizations in preparing for the transition to Post-Quantum Cryptography (PQC)

The following context diagram illustrates the system's position within its environment and its interactions with external entities:



2.2 Product Functions

The Quantum-Proof Systems Scanner provides the following core functionalities.

1. **Asset Input and Discovery**

   The system accepts a manual list of assets provided by the organization, such as domains, URLs, or IP addresses. In addition to manual input, the system performs automated asset discovery to identify additional publicly exposed services using:
   - Certificate-associated domains
   - subdomains
   - Mail and DNS infrastructure hints

   Both manually provided and automatically discovered assets are merged to ensure that no exposed system is missed.

2. **Asset Validation and Scope Enforcement**

   The system verifies that all assets fall within the authorized scanning scope.
   This ensures that the scanner operates within authorized security testing boundaries.

3. **DNS Resolution**

   The system resolves all discovered assets into IP addresses using DNS queries.
   The module performs:
   A record resolution, AAAA record resolution, IP deduplication

4. CDN and WAF Detection

The system identifies whether the target infrastructure is protected by a Content Delivery Network (CDN) or Web Application Firewall (WAF).
If CDN protection is detected, the system marks the TLS configuration as Edge TLS and attempts to identify alternative origins through passive discovery.

5. Service Discovery

The system performs network scanning to identify exposed services on the discovered assets.
Service discovery is conducted using Nmap, targeting commonly used secure service ports
The discovered open ports and services are recorded in the asset database.

6. Protocol Detection

The system analyzes each discovered service to determine the communication protocol used.
Supported protocol detection includes:

- HTTPS and web APIs
- SMTP and STARTTLS email servers
- IMAP and POP3 email retrieval services
- FTPS file transfer services
- SSH remote access services
- VPN endpoints

Each asset is classified according to its service category, such as web, API, mail, file transfer, or VPN.

7. Protocol-Aware Cryptographic Scanning

The system performs protocol-specific cryptographic analysis using specialized scanning modules.
Examples include:

- HTTPS scanning using TestSSL and SSLyze
- SMTP STARTTLS cryptographic inspection
- IMAP and POP3 TLS configuration analysis
- FTPS cipher and TLS inspection
- SSH cryptographic algorithm auditing
- VPN TLS configuration analysis

These modules extract encryption parameters used by each protocol.

8. Cryptographic Data Extraction

The system parses scan results to extract detailed cryptographic parameters, including:

- TLS protocol versions
- Cipher suites
- Key exchange algorithms
- Certificate chains
- Key sizes
- Signature algorithms

This information forms the basis of the organization's cryptographic inventory.

9. Quantum Risk Assessment

The system evaluates extracted cryptographic algorithms to determine their vulnerability to quantum computing attacks.
Algorithms such as:

- RSA
- Elliptic Curve Cryptography (ECC)
- Diffie-Hellman

are flagged as quantum vulnerable.
The system calculates a Harvest-Now-Decrypt-Later (HNDL) risk score for each asset using multiple factors such as algorithm strength, key size, and certificate lifetime.

10. Compliance Mapping

The system maps discovered cryptographic configurations to relevant cybersecurity frameworks and standards.

Findings are correlated with:

- RBI cybersecurity directions
- NIST Post-Quantum Cryptography guidance
- NIST IR 8547 PQC readiness model
- CERT-In security advisories

This enables organizations to assess their regulatory compliance posture.

11. Cryptographic Correlation and Inventory Management

The system aggregates cryptographic findings across all scanned assets to create an organization-wide cryptographic inventory.

Examples include identifying:

- Number of systems using RSA-2048
- Services using outdated TLS versions
- Assets requiring PQC migration

This provides visibility into cryptographic usage across the entire organization.

12. Cryptographic Bill of Materials (CBOM) Generation

The system generates a structured Cryptographic Bill of Materials (CBOM).

The CBOM lists all cryptographic components used by each asset, including:

- Protocol type
- Encryption algorithms
- Key sizes
- Certificates
- Risk classification

The CBOM is exported using the CycloneDX standard format.

13. Scan Comparison and Change Detection

The system compares current scan results with historical scans stored in the database.

This enables detection of:

- Newly introduced vulnerabilities
- Cryptographic upgrades
- Security regressions

This feature allows organizations to track changes in their cryptographic posture over time.

14. PQC Readiness Assessment

The system evaluates the organization's readiness for Post-Quantum Cryptography adoption.

Each asset receives a readiness classification such as:

- Quantum Safe
- Partially Safe
- Vulnerable
- Critical Risk

The system also calculates an organization-level PQC readiness score.

15. Reporting and Visualization

The system provides multiple reporting and visualization capabilities.

Available outputs include:

- Interactive security dashboard
- PDF and HTML security reports
- CBOM export files
- Compliance reports
- Email and webhook alerts

These reports provide actionable insights for security teams and auditors.

16. Continuous Monitoring and Automated Rescanning

2.3User Classes and Characteristics

User Class	Expertise Level	Primary Interaction
Security Analysts	High	Analyze vulnerabilities and cryptographic risks
Compliance Officers	Medium	Review compliance and regulatory reports
System Administrators	High	Implement remediation actions
SOC Teams	High	Monitor alerts and security posture
Organizational Management	Low	View high-level security reports

2.4 Operating Environment

The operating environment for the Quantum-Proof Systems Scanner as listed below.

server system

The system is designed to run on a Linux-based server environment and interacts with external network services during the scanning process.

Operating system

The system is designed to run on a Linux operating system and uses several open-source tools for scanning and analysis.

Recommended operating system:

- Ubuntu Linux (20.04 or later)
- Debian-based Linux distributions

Database:

The system stores scan results and cryptographic inventory data in a database.

Supported database systems include:

- SQLite (for prototype deployments)
- PostgreSQL (for production environments)

Platform

The system operates on a Linux server platform, which provides the necessary environment for running network scanning tools and backend services.

2.5 Design and Implementation Constraints

Technical Constraints (For Deployment)

Network Configuration

- The application must support both public and private network addressing.
- The scanning engine requires outbound internet access to connect to target assets.
- Firewall rules must allow controlled outbound connections to external hosts on relevant ports.
- Network rate limiting must be implemented to prevent excessive traffic during scanning.

Hosting Environment

- The system should support deployment in secure organizational environments, including:
  - On-premise security monitoring servers
  - Virtual machines within enterprise networks
  - Cloud-based infrastructure
- The system should also support restricted environments such as intranet deployments for internal monitoring.
- Containerized deployment using Docker should be supported to simplify installation and portability.

Security Constraints

Access Control

The system must implement Role-Based Access Control (RBAC) to restrict access to authorized users.

User roles may include:

- Security Analyst
- Compliance Officer
- System Administrator
- Monitoring Operator

Data Protection and Encryption

- All communications between the system and the web dashboard must be secured using HTTPS encryption.
- Sensitive scan results stored in the database must be protected using secure storage mechanisms.
- Authentication credentials and API keys must be securely stored and managed.
- The system must only perform scans on authorized targets provided by the organization.

Performance Constraints

The system must be capable of scanning multiple assets without negatively impacting network stability or system performance.

Failover Mechanisms

To ensure reliability, the system must include mechanisms for handling failures.

These include:

- automatic retry for failed scans
- logging and error reporting for failed modules
- database backup and recovery mechanisms

4. User Interface Constraints

4.1 User Experience Consistency

- Uniform layout and menu structure across modules.
- Consistent visual elements such as icons, buttons, and color schemes.
- Standardized navigation workflow for accessing scans, reports, and dashboards.

Compliance and Standards Constraints

- comply with NIST Post-Quantum Cryptography standards
- support NIST IR 8547 PQC readiness assessment models
- provide mappings to relevant regulatory frameworks when generating reports

Reporting Interface Constraints

The system must generate reports in machine-readable and interoperable formats. Supported output formats include:

- JSON
- XML
- CSV
- CycloneDX CBOM format

These formats enable integration with external security tools and compliance systems.

Operational Safety Constraints

- The scanner must operate only on public-facing applications.
- The system must not disrupt live banking services or production infrastructure.
- Scanning must follow controlled rate limits to avoid excessive load on target systems.
- The platform must prevent unauthorized scanning of external infrastructure.

2.6 Assumptions and Dependencies

Assumptions

- Users will access the system through HTML5-compliant web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge.
- Most public-facing services within the organization use TLS-based secure communication protocols (e.g., HTTPS, SMTPS, IMAPS, FTPS).
- The organization provides authorization to scan its public-facing infrastructure.
- The scanning server has stable internet connectivity to access target assets.
- Target infrastructure uses standard networking protocols and ports, enabling automated service detection.

Dependencies

- Database System – PostgreSQL or SQLite for storing assets, scan results, and historical records.
- Security Scanning Tools – Nmap, TestSSL, SSLyze, OpenSSL, and Subfinder for discovery and cryptographic analysis.
- External Data Sources – Certificate Transparency logs, DNS resolution services, and ASN lookup services.
- Post-Quantum Cryptography Standards – Updates from NIST PQC standardization processes for algorithm risk evaluation.
- Network Access – Outbound network access must be available to perform scans on authorized target systems.

Specific Requirements

3.1 Functional Requirements

This section details the functional requirements of the Quantum-Proof Systems, organized by major system features. Each requirement is identified with a unique ID (FR-XX) for traceability and reference.

ID	REQUIREMENT
FR-01	• The system shall accept a list of assets (domains, URLs, or IP addresses) provided by the organization. • The system shall automatically discover additional assets using subdomain enumeration, certificate transparency logs, and DNS records.
FR-02	• The system shall validate that assets fall within the authorized scanning scope. • The system shall remove duplicate assets before scanning.
FR-03	• The system shall resolve domain names into IP addresses using DNS queries. • The system shall support A, AAAA, and CNAME record resolution.
FR-04	• The system shall detect whether an asset is protected by a CDN or WAF. • The system shall label TLS connections as Edge TLS if the asset is protected by CDN infrastructure.
FR-05	• The system shall perform service discovery using Nmap. • The system shall identify open ports related to secure services such as HTTPS, SMTP, IMAPS, FTPS, SSH, and VPN endpoints.
FR-06	• The system shall determine the protocol running on each discovered service. • Supported protocols include HTTPS, SMTP (STARTTLS), IMAP, POP3, FTPS, SSH, and VPN services.
FR-07	• The system shall perform protocol-aware cryptographic scanning using tools such as TestSSL and SSLyze. • The scanner shall extract TLS configurations including cipher suites, TLS versions, certificates, and key exchange algorithms.
FR-08	• The system shall parse scan results to identify cryptographic parameters such as cipher suites, key sizes, certificate chains, and signature algorithms.
FR-09	• The system shall analyze cryptographic algorithms and identify those vulnerable to quantum attacks. • The system shall calculate an HNDL (Harvest Now Decrypt Later) risk score for each asset. • HNDL Risk Score (0–10) = (Algorithm Vulnerability Score × 0.40)+ (Key Size Risk Score × 0.20)+ (Data Sensitivity Weight × 0.30) + (Certificate Expiry Risk × 0.10)
FR-10	• The system shall store discovered cryptographic information in a centralized inventory database. • The system shall maintain historical scan records.
FR-11	• The system shall generate a Cryptographic Bill of Materials (CBOM) listing cryptographic components used by each asset. • The CBOM shall be generated in CycloneDX format.
FR-12	• The system shall map findings to relevant cybersecurity guidelines including NIST PQC standards and CERT-In recommendations.
FR-13	• The system shall evaluate each asset's readiness for Post-Quantum Cryptography. • The system shall assign classifications such as Quantum Safe, Partially Safe, Vulnerable, or Critical.
FR-14	• The system shall support periodic rescanning of assets. • The system shall compare current scan results with historical scans to detect security regressions.
FR-15	• The system shall provide dashboards and reports showing cryptographic inventory, risk scores, and PQC readiness metrics.
FR-16	• The system shall provide automated remediation recommendations for cryptographic vulnerabilities detected during scanning. • Priority Score = Risk Score × Asset Criticality

3.2 External Interface Requirements

3.2.1 User Interfaces

The system shall provide a web-based user interface accessible through standard web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.

The user interface shall allow users to:

- input target assets for scanning
- monitor scan progress
- view discovered assets and services
- review cryptographic risk assessments
- access PQC readiness dashboards
- download reports and CBOM files
- provide remediation steps

The interface shall present information in a clear and structured format using tables, charts, and dashboards.

3.2.2 Hardware Interfaces

The system shall operate on standard server hardware and does not require specialized hardware interfaces.

The scanning server must have:

- internet connectivity
- sufficient CPU and memory resources
- network access to target systems.

3.2.3 Software Interfaces

The system interacts with several external software tools and libraries, including:

- Nmap for service discovery
- TestSSL for TLS analysis
- SSLyze for certificate and TLS scanning
- OpenSSL for protocol inspection
- Subfinder for asset discovery

The backend application integrates these tools through system calls and parses their outputs for analysis.

3.2.4 Communication Interfaces

The system interacts with several external software tools and libraries, including:

- Nmap for service discovery
- TestSSL for TLS analysis
- SSLyze for certificate and TLS scanning
- OpenSSL for protocol inspection
- Subfinder for asset discovery

The backend application integrates these tools through system calls and parses their outputs for analysis.

3.3 System Features

1. *Asset Discovery and Management*

The system automatically identifies public-facing assets belonging to an organization. It combines manually provided asset lists with automated discovery techniques to ensure comprehensive asset coverage.

The system shall:

- accept asset lists provided by the organization
- perform automated asset discovery using subdomain enumeration
- query Certificate Transparency logs to identify domains associated with certificates
- analyze DNS records such as MX and SPF
- merge and deduplicate discovered assets
- validate assets against authorized scanning scope

This feature ensures that no exposed infrastructure is missed during scanning.

2. *Service Discovery and Protocol Identification*

The system identifies active services and determines which protocols are running on discovered assets.

The system shall:

- perform service discovery using Nmap
- identify open ports related to secure services
- determine the protocol running on each open service
- classify services as web servers, APIs, mail servers, file transfer services, or VPN endpoints

3. *Protocol-Aware Cryptographic Scanning*

The system analyzes cryptographic configurations used by each discovered service.

The system shall:

- perform TLS configuration analysis using TestSSL and SSLyze
- inspect certificates and cryptographic parameters
- identify TLS versions and supported cipher suites
- analyze key exchange mechanisms and signature algorithms

4. *Quantum Risk Assessment*

The system evaluates cryptographic configurations to determine their vulnerability to quantum computing attacks.
Functional Behavior
The system shall:

- detect cryptographic algorithms vulnerable to quantum attacks (e.g., RSA, ECC)
- calculate a Harvest-Now-Decrypt-Later (HNDL) risk score
- evaluate cryptographic strength based on algorithm type and key size
- prioritize vulnerabilities based on calculated risk levels

5. *Cryptographic Inventory and Correlation*

The system aggregates cryptographic findings across all scanned assets to build an organization-wide cryptographic inventory.
The system shall:

- store cryptographic data in an inventory database
- track certificates, cipher suites, and algorithms
- correlate cryptographic usage across multiple systems
- generate statistics on algorithm usage across the organization

6. *Compliance Mapping*

The system maps discovered cryptographic configurations to relevant cybersecurity standards and regulatory guidelines.
The system shall:

- compare findings against NIST post-quantum cryptography guidance
- map vulnerabilities to CERT-In security advisories
- support compliance assessment with relevant cybersecurity policies

7. *AI-Assisted Remediation Guidance*

The system provides automated remediation recommendations for detected cryptographic vulnerabilities.
The system shall:

- analyze detected vulnerabilities
- generate recommended configuration improvements
- produce remediation playbooks describing steps required to fix vulnerabilities
- prioritize remediation based on risk severity

8. *Continuous Monitoring*

The system continuously monitors cryptographic configurations through periodic rescanning.
The system shall:

- schedule automated rescans of assets
- compare scan results with historical records
- detect newly introduced vulnerabilities
- identify security regressions

9. *CBOM Generation and Reporting*

The system generates detailed reports describing the cryptographic components used across the organization.
The system shall:

- generate a Cryptographic Bill of Materials (CBOM)
- export CBOM in CycloneDX format
- produce security reports in PDF and HTML formats
- provide interactive dashboards displaying risk metrics

3.4 Non-functional Requirements

3.4.1 Performance Requirements

The system must operate efficiently while scanning multiple public-facing assets without overloading network resources.
The following performance requirements apply:

- The system shall support scanning multiple assets concurrently using controlled parallel scanning.
- The scanning engine shall implement rate limiting to avoid overwhelming target servers.
- The system shall complete scanning of small asset sets within a reasonable time frame.
- The system shall support scheduled and automated scans without manual intervention.
- The system shall store scan results and cryptographic findings in the database with minimal delay.
- The system shall support efficient retrieval of historical scan data for comparison and reporting.

3.4.2 Software Quality Attributes

The system must maintain high standards of reliability, security, and maintainability.

Reliability

- The system shall log all scanning operations and system events.
- The system shall retry failed scans when temporary network issues occur.
- The database shall store historical scan results to ensure persistence of data.

Security

- Access to the system shall be restricted using Role-Based Access Control (RBAC).
- All web interface communication shall be secured using HTTPS encryption.
- Only authorized targets within the defined scope shall be scanned.

Maintainability

- The system architecture shall be modular to allow easy updates to scanning modules.
- New protocol scanners or cryptographic analysis modules can be added without major system redesign.

Scalability

- The system shall support scanning of multiple domains and assets.
- The system shall support expansion of the asset inventory database as the number of scanned assets increases.

Usability

- The system dashboard shall provide clear visualizations of cryptographic risks.
- Reports shall present information in a structured and understandable format.

3.4.3 Other Non-Functional Requirements

Additional constraints related to operation and compliance include:

Compliance

- The system shall align with recognized cybersecurity standards and post-quantum cryptography guidelines.

Interoperability

- The system shall generate reports in machine-readable formats, including JSON, XML, and CSV.
- The system shall support CycloneDX CBOM format for cryptographic inventory reporting.

Operational Safety

- The scanner shall operate only on public-facing applications and authorized assets.
- The scanning process must not disrupt normal operation of banking services.

Monitoring and Logging

- The system shall maintain logs of scanning activities, detected vulnerabilities, and system events.
- Logs shall be accessible for auditing and troubleshooting purposes.

4. Technological Requirements

4.1 Technologies used in development of the web application

The system is developed using modern web and backend technologies that support scalable security scanning and data analysis.

The primary technologies include:

Backend Technologies

- Python – Used for implementing the scanning engine, cryptographic analysis modules, and risk assessment algorithms.
- FastAPI / Flask – Used to build REST APIs for communication between system modules and the dashboard.

Frontend Technologies

- HTML5 – Structure of the web interface.
- CSS3 – Styling and layout of the dashboard.
- JavaScript – Interactive user interface behavior.
- React.js – Used for building dynamic dashboard components.

Security Scanning Tools

The application integrates several external security tools to perform scanning and cryptographic analysis:

- Nmap – Network service discovery and port scanning.
- TestSSL – TLS configuration analysis.
- SSLyze – TLS and certificate inspection.
- OpenSSL – Protocol-level cryptographic inspection.
- Subfinder – Asset discovery through subdomain enumeration.

4.2 I.D.E. (Integrated Development Environment)

- Visual Studio Code (VS Code) – Primary IDE used for backend and frontend development.
- PyCharm – Used for Python development and debugging.
- Git – Version control system used for managing source code.

4.3 Database Management Software

Supported database systems include:

SQLite

- Used for prototype or small-scale deployments.
- Lightweight and easy to deploy.

PostgreSQL

- Used for production deployments.
- Supports large datasets and concurrent access.
- Suitable for storing asset inventories, scan results, and historical data.

5. Security Requirements

5.1 Access Control

The system shall implement Role-Based Access Control (RBAC) to restrict access to authorized users.

5.2 Authentication and Authorization

The system shall require secure authentication for all users accessing the dashboard.
Authentication mechanisms may include:

- username and password authentication
- secure session management
- optional multi-factor authentication for administrators

5.3 Secure Communication

All communication between users and the system must be protected using HTTPS encryption.
Security requirements include:

- TLS encryption for all web dashboard communication
- secure API communication between system components
- protection against man-in-the-middle attacks

5.4 Authorized Scanning

The system must enforce strict controls to ensure that scanning operations are performed only on authorized assets.
Security controls include:

- validation of target domains and IP addresses
- allowlist-based scanning scope
- prevention of unauthorized or external target scanning

5.5 Data Protection

The system shall protect stored data and sensitive information.
Security measures include:

- secure storage of scan results and asset inventories
- restricted access to database records
- protection of API keys and credentials

5.6 Logging and Audit Trails

The system shall maintain detailed logs of all system operations.
Logs shall record:

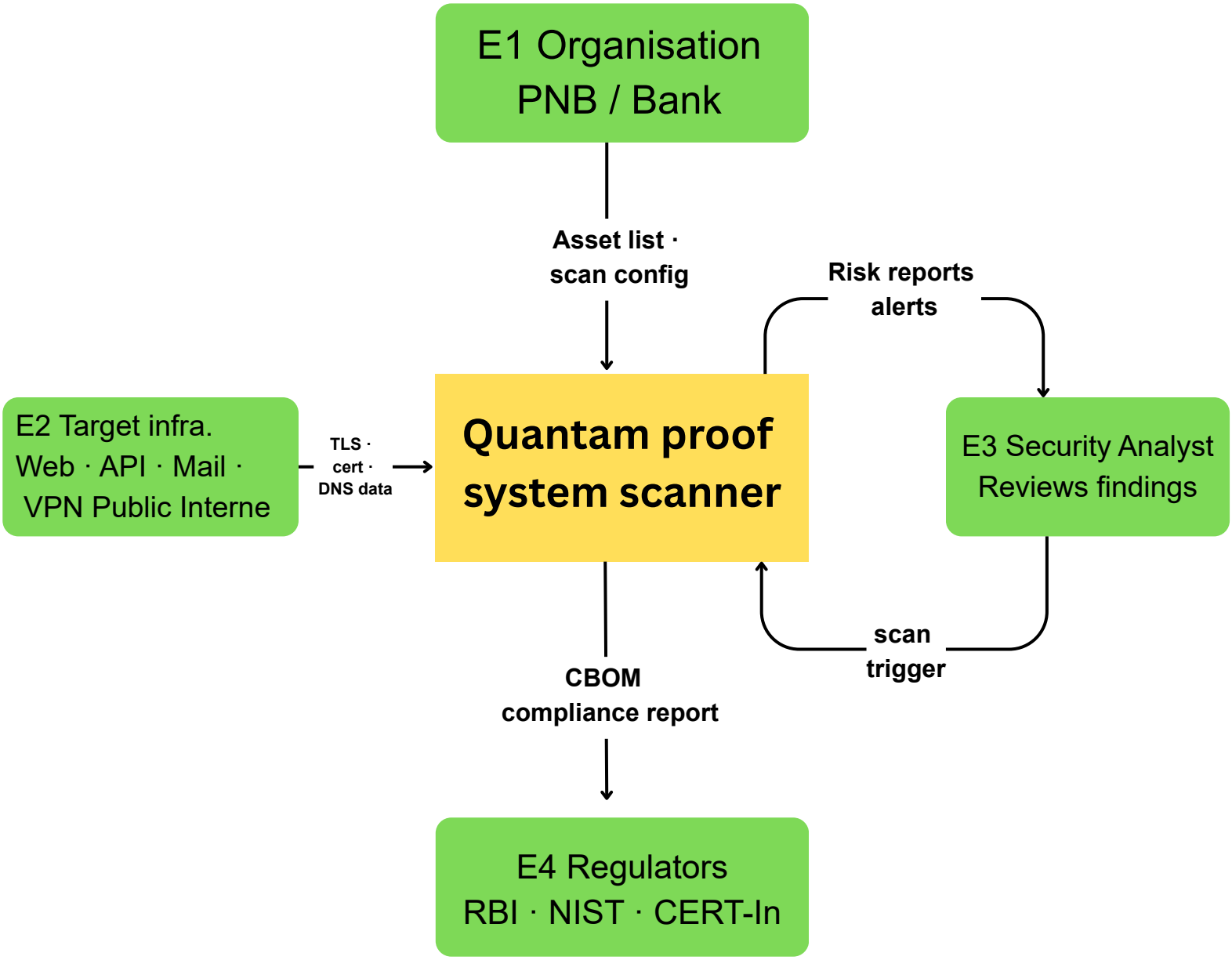
- user login activities
- scan initiation and completion
- detected vulnerabilities
- system errors and alerts

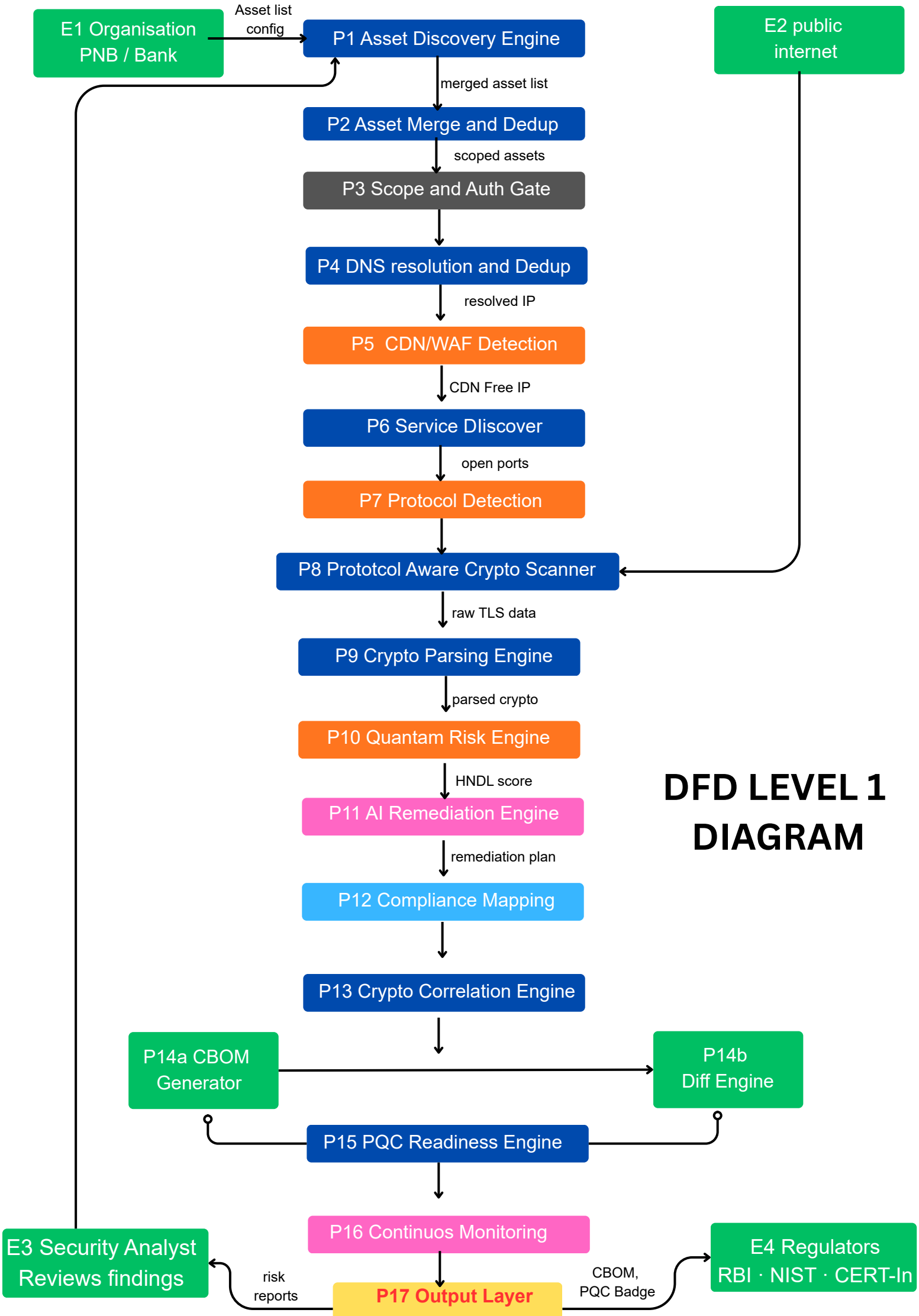
5.7 System Integrity Protection

The system shall ensure that scanning modules and analysis components cannot be tampered with by unauthorized users.
Measures include:

- restricted access to system configuration files
- secure update mechanisms for scanning tools
- validation of system modules before execution

DFD LEVEL 0 DIAGRAM




E1 Organisation PNB / Bank
Asset list config
P1 Asset Discovery Engine
E2 public internet
merged asset list
P2 Asset Merge and Dedup
scoped assets
P3 Scope and Auth Gate
P4 DNS resolution and Dedup
resolved IP
P5 CDN/WAF Detection
CDN Free IP
P6 Service DIiscover
open ports
P7 Protocol Detection
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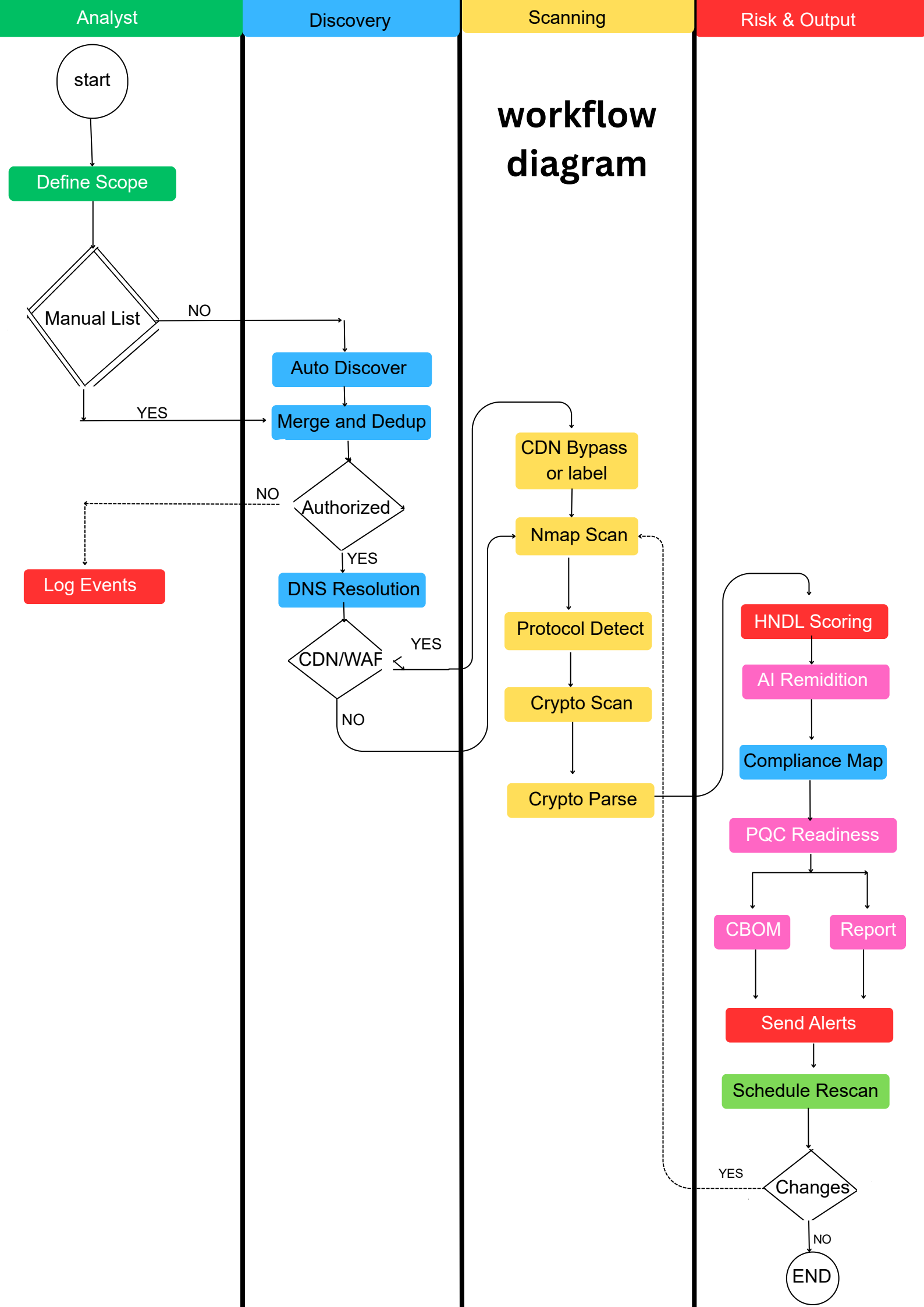
raw TLS data
P9 Crypto Parsing Engine
parsed crypto
P10 Quantam Risk Engine
HNDL score
P11 AI Remediation Engine
remediation plan
P12 Compliance Mapping
P13 Crypto Correlation Engine
P14a CBOM Generator
P14b Diff Engine
P15 PQC Readiness Engine
P16 Continuos Monitoring
E3 Security Analyst Reviews findings
risk reports
P17 Output Layer
CBOM, PQC Badge
E4 Regulators RBI · NIST · CERT-In
DFD LEVEL 1 DIAGRAM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NO
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Nmap Scan
Protocol Detect
Crypto Scan
Crypto Parse
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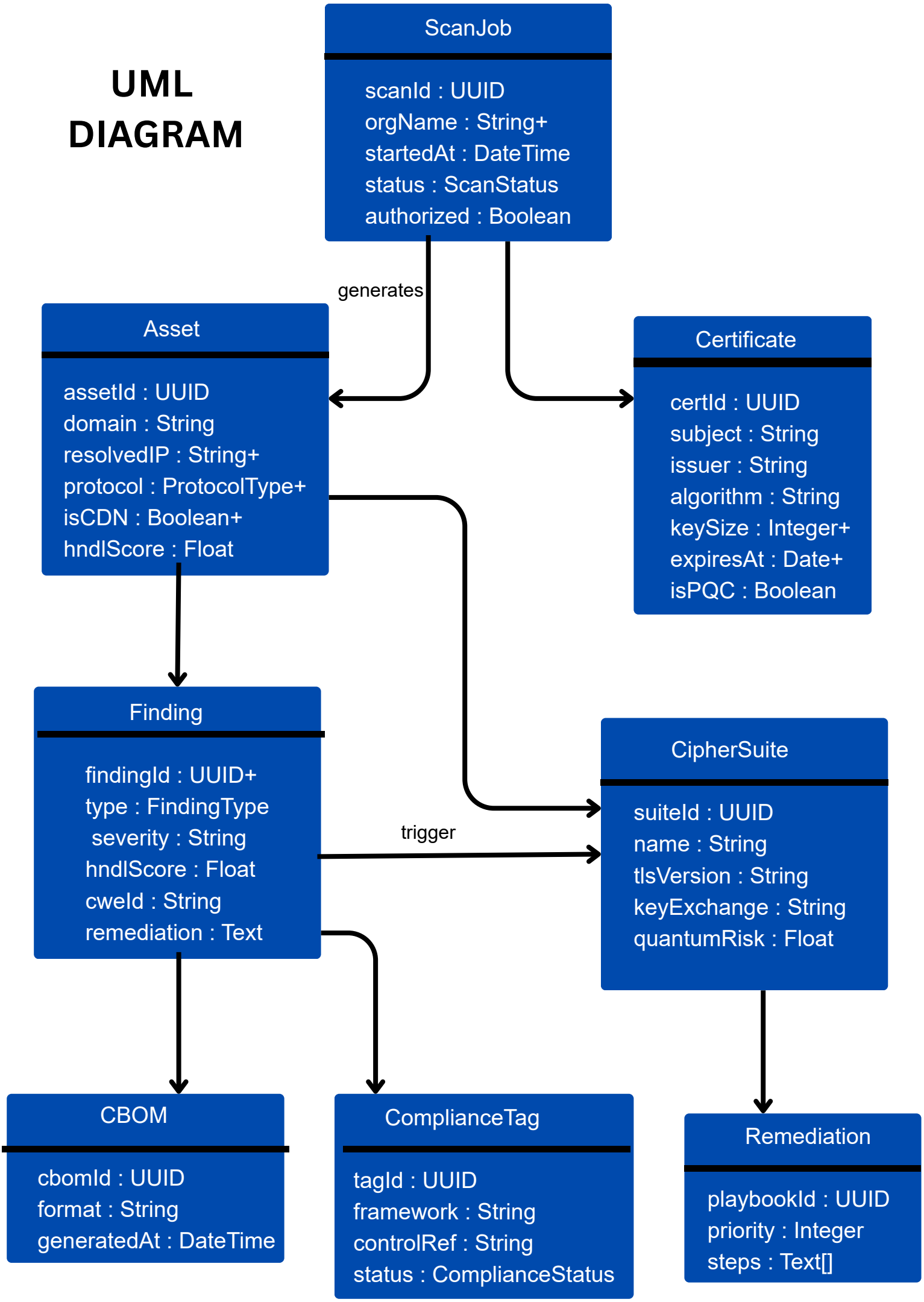
Changes
NO
END

UML DIAGRAM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isCDN : Boolean+
hndlScore : Float
Certificate
certId : UUID
subject : String
issuer : String
algorithm : String
keySize : Integer+
expiresAt : Date+
isPQC : Boolean
Finding
findingId : UUID+
type : FindingType
severity : String
hndlScore : Float
cweId : String
remediation : Text
trigger
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Table 9 : Minimum Elements pertaining to Cryptographic Asset

Cryptographic Asset Type	Element	Description
Algorithms	Name	The name of the cryptographic algorithm or asset. For example, "AES-128-GCM" refers to the AES algorithm with a 128-bit key in Galois/Counter Mode (GCM).
	Asset Type	Specifies the type of cryptographic asset. For algorithms, the asset type is "algorithm".
	Primitive	Describes the cryptographic primitive. For "SHA512withRSA", the primitive is "signature" as it's used for digital signing.
	Mode	The operational mode used by the algorithm. For example, "gcm" refers to the Galois/Counter Mode used with AES encryption.

Cryptographic Asset Type	Element	Description
	Crypto Functions	The cryptographic functions supported by the asset. For example, the functions in the case of "AES-128-GCM" are key generation, encryption, decryption, and authentication tag generation.
	Classical security level	The classical security level represents the strength of the cryptographic asset in terms of its resistance to attacks using classical (non-quantum) methods. For AES-128, it's 128 bits.
	OID	The Object Identifier (OID) is a globally unique identifier used to refer to the algorithm. It helps in distinguishing algorithms across different systems. For example, "2.16.840.1.101.3.4.1.6" for AES-128-GCM, "1.2.840.113549.1.1.13" for SHA512withRSA
	List	Lists the cryptographic algorithms employed by the quantum device or system, allowing for an assessment of its security capabilities, especially in the context of post-quantum encryption standards.
Keys	Name	The name of the key, which is a unique identifier for the key used in cryptographic operations.
	Asset Type	Defines the type of cryptographic asset. For keys, the asset type is typically "key".
	id	A unique identifier for the key, such as a key ID or reference number.
	state	The state of the key, such as whether it is active, revoked, or expired.
	size	The size of the key, typically measured in bits. For example, a 128-bit key or a 2048-bit RSA key.
	Creation Date	The date when the key was created.
	Activation Date	The date when the key became operational or was first used.
Protocols	Name	The name of the cryptographic protocol, such as TLS, IPsec, or SSH

Cryptographic Asset Type	Element	Description
	Asset Type	Defines the type of cryptographic asset. In this case, it would be a "protocol"
	Version	The version of the protocol used, such as TLS 1.2 or TLS 1.3.
	Cipher Suites	The set of cryptographic algorithms and parameters supported by the protocol for tasks like encryption, key exchange, and integrity checking.
	OID	The Object Identifier (OID) associated with the protocol, identifying its unique specifications.
Certificates	Name	The name of the certificate, typically referring to its subject or the entity it represents (e.g., a website).
	Asset Type	Defines the type of cryptographic asset. For certificates, the asset type is "certificate".
	Subject Name	This refers to the Distinguished Name (DN) of the entity that the certificate represents. It typically contains information about the organization, domain name
	Issuer Name	The issuer is the Certificate Authority (CA) that issued and signed the certificate. This field contains the DN of the CA that verified and issued the certificate.
	Not Valid Before	This specifies the date and time from which the certificate is valid.
	Not Valid After	This specifies the expiration date and time of the certificate. The certificate becomes invalid after this timestamp.
	Signature Algorithm Reference	This refers to the cryptographic algorithm used to sign the certificate. It provides a reference to the algorithm and its OID (Object Identifier).
	Subject Public Key Reference	This points to the public key used by the subject (the entity being identified in the certificate). It provides a reference to the key's details, including the algorithm.
	Certificate Format	Specifies the format of the certificate. Common formats include X.509, which is the most widely used format for certificates.

Cryptographic Asset Type	Element	Description
	Certificate Extension	This refers to the file extension associated with the certificate. It is commonly .crt for certificates in the X.509 format.